WORKED SOLUTION

STEP III 2012, Question 9

Mechanics • Rotating Pulley Dynamics

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Read the question before you go on. It is in the STEP Support Programme's question database:

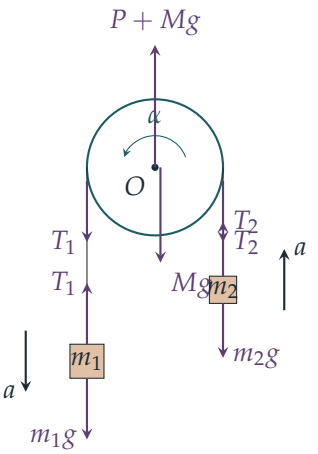
step.maths.org/questions/12-s3-q9

The key idea

The measured force on the axle is the clue. The centre of mass of the pulley doesn't move, so the axle must hold up the pulley's weight and also balance the two string tensions. That makes P the sum of the tensions, and from that the acceleration of the particles follows. Once we know the acceleration, the pulley's rotational equation gives the moment of inertia. A friction couple changes that one equation and nothing else.

Solution

Model. The particles are point masses and the string is light and inextensible. The string doesn't slip on the pulley, so its speed equals the rim speed of the disc. We take the system to have been released from rest, so the heavier particle m_1 descends. Let a be the acceleration of m_1 downwards, which is also the acceleration of m_2 upwards. The pulley's angular acceleration is then $\alpha = a/r$. The tension is T_1 in the string on the m_1 side and T_2 on the m_2 side (the two differ because the pulley has inertia). Let I be the moment of inertia about the axle.



Newton's second law for each particle gives

$$m_1 g - T_1 = m_1 a, \qquad T_2 - m_2 g = m_2 a,$$

so

$$T_1 = m_1(g - a), \qquad T_2 = m_2(g + a).$$

The axle force. The centre of mass of the pulley is at O, which is fixed, so the net force on the pulley is zero. The forces acting on it are its weight Mg, the two tensions (both pulling down, since the string hangs vertically on each side) and the force from the axle. A friction couple (which we add in part (ii)) has no resultant force, so it doesn't enter here. The vertical force on the axle is therefore $Mg + T_1 + T_2$, and comparing with the measurement gives

$$P = T_1 + T_2 = (m_1 + m_2)g - (m_1 - m_2)a,$$

so

$$a = \frac{(m_1 + m_2)g - P}{m_1 - m_2}. \tag{1}$$

This uses only the measurement and the particles' equations, so it holds in both parts.

Part (i)

With no friction at the axle, only the tensions have moments about O. Taking moments about the axle,

$$(T_1 - T_2)r = I\alpha = \frac{Ia}{r}, \qquad \text{so} \qquad I = \frac{(T_1 - T_2)r^2}{a}.$$

Now $T_1 - T_2 = (m_1 - m_2)g - (m_1 + m_2)a$, so

$$I = r^2 \left(\frac{(m_1 - m_2)g}{a} - (m_1 + m_2) \right).$$

Substituting (1) for a:

$$\begin{aligned} I &= r^2\left(\frac{(m_1-m_2)^2g}{(m_1+m_2)g-P}-(m_1+m_2)\right)\\ &= r^2\,\frac{(m_1-m_2)^2g-(m_1+m_2)^2g+(m_1+m_2)P}{(m_1+m_2)g-P}\\ &= \frac{((m_1+m_2)P-4m_1m_2g)r^2}{(m_1+m_2)g-P}, \end{aligned}$$

using $(m_1+m_2)^2-(m_1-m_2)^2=4m_1m_2$. This is $(*)$, as required.

Part (ii)

Call the new calculated value I'. The friction couple C opposes the rotation, and the rotation is in the sense driven by T_1, so the moment equation becomes

$$(T_1-T_2)r-C=\frac{I'a}{r}.$$

The acceleration is still given by (1), because the same measured P is used and the axle-force argument doesn't involve C. So T_1-T_2 and a are exactly as in part (i), and

$$I'=\frac{(T_1-T_2)r^2}{a}-\frac{Cr}{a}=(*)-\frac{Cr}{a}.$$

Ask yourself How do we know the sign of a?

The system starts from rest and the pulley turns with m_1 going down, so m_1 must have a positive downward acceleration. (If a were zero or negative, m_1 would never start moving downwards.) So $a>0$. Since $C>0$ and $r>0$, we get $Cr/a>0$, and **the value calculated with friction is smaller than** $(*)$. This makes sense: part of the net torque from the tensions goes into overcoming friction, so less inertia is needed to explain the same observed acceleration.

The bound on C. Substitute the tensions into the moment equation:

$$((m_1-m_2)g-(m_1+m_2)a)r-C=\frac{I'a}{r},$$

which rearranges to

$$(m_1-m_2)rg-C=a\left(\frac{I'}{r}+(m_1+m_2)r\right).$$

On the right, $a>0$, $I'\geq 0$ (a moment of inertia can't be negative) and $(m_1+m_2)r>0$, so the right-hand side is strictly positive. Hence

$$C<(m_1-m_2)rg,$$

as required. In words, the friction couple must be less than the couple $(m_1-m_2)gr$ from the out-of-balance weight. If it weren't, the system wouldn't start to move at all.

Checks

- *Light pulley:* if $I = 0$ and $C = 0$ then $a = \frac{(m_1 - m_2)g}{m_1 + m_2}$ (the familiar Atwood result) and $P = \frac{4m_1 m_2 g}{m_1 + m_2}$. The numerator of $(*)$ then vanishes, as it should.
- *Stationary pulley:* as $a \to 0$ we get $P \to (m_1 + m_2)g$, the denominator of $(*)$ tends to zero and $(*)$ blows up. An infinitely heavy pulley doesn't accelerate.
- *Numerically:* with $g = 9.8$, $m_1 = 3$, $m_2 = 1$, $r = 0.5$, $I = 0.7$ and $C = 2$ (SI units), the true motion has $a \approx 2.29\ \mathrm{m\,s^{-2}}$ and $P \approx 34.6$ N. Formula $(*)$ gives $I \approx 1.14$, which is too big by exactly $Cr/a \approx 0.44$, while the friction model gives back 0.7. Here $C = 2 < (m_1 - m_2)rg = 9.8$. We tested 10^4 random cases the same way.

Common slips

- Taking the tension to be the same on both sides. That holds only for a light or frictionless pulley, and here it would make I disappear from the problem.
- Forgetting the pulley's own weight in the axle force. That is why the measurement is written as $P + Mg$.
- Thinking the friction couple changes the axle force. A couple has zero resultant, so $P = T_1 + T_2$ in both parts.

Beyond STEP

This is a small example of *inferring a hidden parameter from a measurement*. The answer depends on the model you assume, and leaving out friction biases the estimate in a predictable direction. In first-year mechanics, rigid bodies are treated with the full apparatus of linear and angular momentum, and the moment of inertia becomes a tensor for bodies rotating about general axes. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers fixed-axis rotation and Atwood machines with massive pulleys in depth.

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